Multi-Task Financial NLP: Stance & Sentiment Classification

COMP6713 — Natural Language Processing — 2026 T1

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Problem Statement & Datasets

Problem statement:

Given a short financial sentence, predict *both* its monetary-policy stance and its market sentiment under heavy class imbalance and limited labelled data ($<$2,500 per task).

Two 3-class classification tasks:

- **Stance** (FOMC monetary policy) {hawkish, dovish, neutral}
- **Sentiment** (financial news) {positive, negative, neutral}

Datasets (70/10/20 stratified, seed 42):

	Total	Train	Test
FOMC [1]	2,480	1,736	496
FPB all-agree [2]	2,264	1,584	453

- FPB test: 61/278/114 (**neu-dominant**)
- FOMC test: 130/121/245 (balanced-ish)
- LM lexicon [3] for rule-based baseline

Baselines & Pre-trained Probes

1. LM Lexicon (rule-based) [3]

- **5 curated word lists**: LM-positive ($\sim$110), negative ($\sim$230), uncertainty ($\sim$90), hawkish ($\sim$40), dovish ($\sim$50).
- **No training**: sign-count the words; majority wins.
- Quick, interpretable floor.

Sentiment F1 $\approx$ 0.53*; Stance F1* $\approx$ 0.39.

2. TF-IDF linear models

- **Unigram+bigram** (`max_features=50k`, `sublinear_tf`) + LR.
- **LinearSVC** (C=1, class-weighted).
- **Trigrams** (`ngram=(1,3)`, 80k, `min_df=2`) + LR.
- **Hybrid**: TF-IDF + 8 scaled LM-lexicon features.

Best: TF-IDF+SVM — F1 0.85 *sent,* 0.61 *stance.*

3. Zero-/Few-shot probes

- **FinBERT zero-shot**: native 3-class head with a positive$\rightarrow$hawkish / negative$\rightarrow$dovish proxy for stance.
- **Few-shot** k=16: freeze encoder, train `Linear(768,3)` on 48 examples (Adam lr 10^{-3}, 200 ep).
- Probed encoders: **FinBERT**, **BERT-base**, **RoBERTa-base**.

FinBERT few-shot: F1 0.967 *sent —*

Methodology: 6-Stage Model Ladder

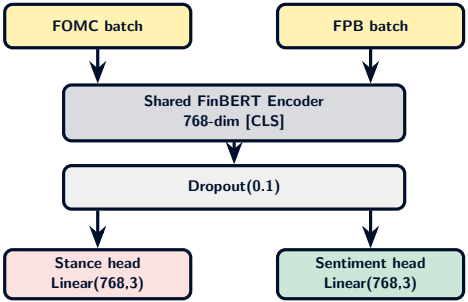
FinBERT fine-tune

- AdamW, lr $2\mathrm{e}{-5}$, 5 epochs, batch 32
- Class weights $(1.272, 1.365, 0.675)$ for stance
- Linear warmup (10%) $\rightarrow$ linear decay

BERT-base LLRD + gradual unfreezing [4, 5]

- $\eta_d = 2\mathrm{e}{-5} \cdot 0.9^{\,d}$ per layer group
- Unfreeze 1 layer group per epoch (10 epochs)
- Label smoothing $\varepsilon = 0.1$

Multi-task FinBERT Architecture (main contribution)



Training [6]:

- **Alternating batches**: one FOMC, then one FPB; active task's head receives gradient, inactive receives zero.
- AdamW lr $2e{-}5$, 8 epochs, batch 32.
- Per-task CE (class-weighted for stance).
- Best checkpoint by *average* validation macro-F1.

Shared encoder $\Rightarrow$ each task regularises the other $\Rightarrow$ free data-augmentation effect.

Results — All 12 Systems

Model	Sent. Acc	Sent. F1	Stance Acc	Stance F1
LM Lexicon (rule-based)	0.6932	0.5315	0.4153	0.3885
TF-IDF + LR	0.8720	0.8232	0.6089	0.5873
TF-IDF + SVM (LinearSVC)	0.8940	0.8534	0.6331	0.6061
TF-IDF (trigrams) + LR	0.8786	0.8310	0.6109	0.5914
TF-IDF + LM Lexicon (hybrid)	0.8543	0.8050	0.6109	0.5863
FinBERT zero-shot (native)	0.9735	0.9650	0.4980	0.4874
FinBERT few-shot (k=16 probe)	0.9779	0.9670	0.4859	0.4534
BERT-base few-shot (k=16)	0.7417	0.6500	0.3851	0.3744
RoBERTa-base few-shot (k=16)	0.7682	0.6722	0.3730	0.3600
FinBERT (fine-tuned)	0.9669	0.9459	0.6371	0.6194
BERT-base LLRD + Gradual UF	0.9757	**0.9670**	**0.6512**	**0.6383**
Multi-task FinBERT	**0.9779**	**0.9666**	0.6492	**0.6384**

- **Best non-neural**: TF-IDF + SVM — sub-ms inference; defensible floor for latency-sensitive / air-gapped deployment.
- **Best without fine-tuning**: FinBERT few-shot (k=16, 48 examples) — matches full fine-tune sentiment (0.9670 vs 0.9459). Domain pretraining dominates in low-data.
- **Best overall**: Multi-task FinBERT ≡ BERT-LLRD (within ±0.1 pp). Multi-task **beats** single-task

Error Analysis — Multi-task FinBERT

Stance (FOMC test, N=496)

Class	Per-class F1
dovish	0.6174
hawkish	0.5869
neutral	**0.7109**

- Most errors sit on the **neutral vs. leaning** boundary — hedged language ("*appropriate to maintain*") is genuinely ambiguous.
- Hawkish is the weakest class (lowest count in train and most lexically diverse).

Sentiment (FPB test, N=453)

Class	Per-class F1
negative	0.9500
neutral	**0.9928**
positive	0.9569

- Only ∼10 **errors total** across 453 test sentences — sentiment is near-saturated.
- Remaining errors are subtle *comparative* constructions ("*profit fell less than expected*").

Deployment & Live Demo

Three interfaces:

- **CLI** `cli.py`:
  `--text` / `--file` / interactive
- **Gradio demo** `demo.py`:
  web UI at `localhost:7860`
- **HuggingFace Hub**: 5 public repos
  `Louisnguyen/multitask-finbert-financial`
  `Louisnguyen/finbert-financial-{stance,sentiment}`
  `Louisnguyen/bert-llrd-financial-{stance,sentiment}`

Live Demo

Switching to Gradio at `http://localhost:7860`

Example inputs:

- "The Fed signaled further rate hikes ahead."
- "Operating profit rose sharply in Q3."
- "Some participants observed inflation pressures have eased."

Summary & Takeaways

Problem

Stance (FOMC) + sentiment (FPB) classification — 2,480 + 2,264 sentences.

Approach

12-system ladder: lexicon → TF-IDF → zero/few-shot → fine-tune → **LLRD + gradual unfreeze** → **multi-task FinBERT**.

Headline result

Multi-task FinBERT: **97.79%** / **0.9666** sentiment, **64.92%** / **0.6384** stance. Multi-task beats single-task FinBERT by **+2.07 pp** sent, **+1.90 pp** stance.

Lesson

Domain pretraining **and** careful fine-tuning recipes are *substitutes*, not complements — BERT-LLRD ties Multi-task FinBERT.

Thank you for listening!

Q&A time

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